

Aitezaz Sikandar Khan

Full-Stack Developer

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Professional Summary

Full-Stack Software Engineer specializing in building web applications, real-time architectures, and cloud-deployed services using TypeScript, React, Next.js, and Node.js. Practical experience in MongoDB schema design, REST API engineering, WebSocket communication (Socket.io), role-based access control systems, and containerized deployments with Docker and AWS EC2.

Education

University of Peshawar

2022 -- 2026

Bachelor of Science in Computer Science (GPA: 3.60 / 4.00)

- **Achievements:** Secured the 95th percentile nationwide in the HEC National Skills Competency Test (NSCT) organized by HEC, PSEB, and P@SHA (April 2026).
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering

Technical Skills

Languages: JavaScript, TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI (MUI), Responsive Design

Backend: Node.js, Express.js, RESTful APIs, REST API Design, WebSockets (Socket.io)

Databases: MongoDB, Mongoose ODM, MySQL, Firebase

DevOps & CI/CD: Docker, GitHub Actions, AWS EC2, Nginx, Linux

Tools & Practices: Git, GitHub, Postman, Jest, Zod, Stripe API, Cloudinary, Agile/Scrum

Professional Experience

e-strats

Islamabad, Pakistan

Software Engineer Intern (MERN Stack)

Jun 2026 -- Aug 2026

- Developed RESTful API endpoints in Node.js and Express for a cloud-based access control system to manage door controllers, digital credentials, and schedule-based zone permissions.
- Built real-time monitoring dashboards in React.js using Socket.io to stream live door unlock events, entry/exit logs, and unauthorized access alerts.
- Implemented role-based access control (RBAC) middleware with JWT authentication and Bcrypt hashing to secure administrative routes and ensure multi-tenant data isolation in MongoDB.

Full-Stack Developer

Freelance / Client Projects

2024 -- Present

- Delivered custom MERN-stack web applications for clients, including a complaint management portal featuring automated email notifications (Nodemailer) and media uploads (Cloudinary).
- Structured MongoDB collections with indexing and validation schemas to optimize query performance and maintain data integrity across API endpoints.

Projects

Collaborative Study Platform | React, Socket.io, Node.js, Gemini & Groq

- Developed a real-time virtual classroom utilizing Socket.io to synchronize slide changes, active users, and typing indicators with sub-100ms latency for up to 50 concurrent students.
- Designed a multi-LLM fallback mechanism that automatically redirects requests from Groq (Llama 3) to Google Gemini (3.5-Flash) when rate limits are hit, preventing AI chat interruptions.
- Eliminated audio latency by streaming text-to-speech voice narration directly to the browser using Express chunked transfer encoding headers.

Multi-Role Clinic Management System | React, Node.js, Express, MongoDB, Gemini API

- Built a multi-role portal (Admin, Doctor, Patient) with appointment booking, using moment.js validation to prevent scheduling conflicts and Nodemailer for automated email reminders.
- Integrated an AI symptom-analysis assistant via the Gemini API, structuring prompt fallbacks to summarize patient history and generate relevant questions for doctor visits.
- Designed MongoDB aggregation pipelines to calculate monthly appointment distributions and doctor earnings, rendering the data on interactive dashboard charts.

Full-Stack E-Commerce | Next.js, TypeScript, Redux, Stripe API, MongoDB

- Developed a typed storefront using Next.js App Router and TypeScript, managing persistent shopping cart states across pages with Redux Toolkit.
- Integrated Stripe Checkout for payments, implementing a backend webhook route to save orders to MongoDB and reset customer carts upon payment verification.
- Secured checkout forms and API endpoints against invalid data using schema-based Zod validation and React Hook Form.